

A Technical Introduction to Convolutional neural network

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

Section 1

Video analysis Compared to image data domains, there is relatively little work on applying CNNs to video classification. Video is more complex than images since it has another (temporal) dimension. However, some extensions of CNNs into the video domain have been explored. One approach is to treat space and time as equivalent dimensions of the input and perform convolutions in both time and space. Another way is to fuse the features of two convolutional neural networks, one for the spatial and one for the temporal stream. Long short-term memory (LSTM) recurrent units are typically incorporated after the CNN to account for inter-frame or inter-clip dependencies. Unsupervised learning schemes for training spatio-temporal features have been introduced, based on convolutional gated restricted Boltzmann machines and independent subspace analysis. Its application can be seen in text-to-video models.

Section 2

Human interpretable explanations End-to-end training and prediction are common practice in computer vision. However, human interpretable explanations are required for critical systems such as self-driving cars. With recent advances in visual salience, spatial attention, and temporal attention, the most critical spatial regions/temporal instants could be visualized to justify the CNN predictions.

Section 3

Fully connected layers Fully connected layers connect every neuron in one layer to every neuron in another layer. It is the same as a traditional multilayer perceptron neural network (MLP). Each neuron in the fully connected layer receives input from all the neurons in the previous layer. These inputs are weighted and summed with the corresponding biases, and then passed through an activation function to perform a nonlinear transformation, generating the output. The flattened matrix goes through a fully connected layer to classify the images.

Section 4

Translation equivariance and aliasing It is commonly assumed that CNNs are invariant to shifts of the input. Convolution or pooling layers within a CNN that do not have a stride greater than one are indeed equivariant to translations of the input. However, layers with a stride

greater than one ignore the Nyquist–Shannon sampling theorem and might lead to aliasing of the input signal. While, in principle, CNNs are capable of implementing anti-aliasing filters, it has been observed that this does not happen in practice, and therefore yield models that are not equivariant to translations. Furthermore, if a CNN makes use of fully connected layers, translation equivariance does not imply translation invariance, as the fully connected layers are not invariant to shifts of the input. One solution for complete translation invariance is avoiding any down-sampling throughout the network and applying global average pooling at the last layer. Additionally, several other partial solutions have been proposed, such as anti-aliasing before downsampling operations, spatial transformer networks, data augmentation, subsampling combined with pooling, and capsule neural networks.

Section 5

Number of filters Since feature map size decreases with depth, layers near the input layer tend to have fewer filters while higher layers can have more. To equalize computation at each layer, the product of feature values with pixel position is kept roughly constant across layers. Preserving more information about the input would require keeping the total number of activations (number of feature maps times number of pixel positions) non-decreasing from one layer to the next. The number of feature maps directly controls the capacity and depends on the number of available examples and task complexity.

Section 6

When dealing with high-dimensional inputs such as images, it is impractical to connect neurons to all neurons in the previous volume because such a network architecture does not take the spatial structure of the data into account. Convolutional networks exploit spatially local correlation by enforcing a sparse local connectivity pattern between neurons of adjacent layers: each neuron is connected to only a small region of the input volume. The extent of this connectivity is a hyperparameter called the receptive field of the neuron. The connections are local in space (along width and height), but always extend along the entire depth of the input volume. Such an architecture ensures that the learned filters produce the strongest response to a spatially local input pattern.

Section 7

Because the degree of model overfitting is determined by both its power and the amount of training it receives, providing a convolutional network with more training examples can reduce overfitting. Because there is often not enough available data to train, especially considering that some part should be spared for later testing, two approaches are to either generate new data from scratch (if possible) or perturb existing data to create new ones. The

latter one is used since mid-1990s. For example, input images can be cropped, rotated, or rescaled to create new examples with the same labels as the original training set.

Section 8

For example, in CIFAR-10, images are only of size $32 \times 32 \times 3$ (32 wide, 32 high, 3 color channels), so a single fully connected neuron in the first hidden layer of a regular neural network would have $32 \times 32 \times 3 = 3,072$ weights. A 200×200 image, however, would lead to neurons that have $200 \times 200 \times 3 = 120,000$ weights. Also, such network architecture does not take into account the spatial structure of data, treating input pixels which are far apart in the same way as pixels that are close together. This ignores locality of reference in data with a grid-topology (such as images), both computationally and semantically. Thus, full connectivity of neurons is wasteful for purposes such as image recognition that are dominated by spatially local input patterns. Convolutional neural networks are variants of multilayer perceptrons, designed to emulate the behavior of a visual cortex. These models mitigate the challenges posed by the MLP architecture by exploiting the strong spatially local correlation present in natural images. As opposed to MLPs, CNNs have the following distinguishing features:

Section 9

Animal behavior detection CNNs have been applied in ecological and behavioral research to automatically detect and quantify animal behavior from visual data, enabling identification of animals, tracking of individuals, estimation of pose, and classification of specific actions such as feeding, and social interactions. Combined with multi-object tracking and temporal modeling, these systems can extract behavioral sequences over extended recordings, reducing reliance on manual annotation and increasing throughput for studies of individual variation, social networks, and collective dynamics.